

Estimation and Test of Hypothesis

Subject / Theme :

Date :

☐ Sat ☐ Sun ☐ Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri

Hypothesis:-

A statement about the nature of a population

Example: Students who eat breakfast will perform better on a stat exam than students who do not eat breakfast

Test of Hypothesis:-

The statistical procedure which is used to verify any statement or assumption about population parameter on the basis of sample observations is known as test of significance.

Null Hypothesis:- H_0 is the research hypothesis

The hypothesis which we are going to test for possible rejection under the assumption that it is true.

Null Hypothesis is denoted by H_0 .

$$H_0 = \mu_1 = \mu_2 \rightarrow \mu = \text{mean}$$

Alternative Hypothesis:-

Each of all possible hypothesis other than null hypothesis is called alternative hypothesis and is usually denoted by H_1

$$H_1, H_0: \mu_1 \neq \mu_2$$

Subject / Theme:

 $\alpha \uparrow, \beta \downarrow$ $\alpha \downarrow, \beta \uparrow$

☐ Type-I error:- [False Positive]

The error of rejecting H_0 (accepting H_1) which is true is called Type-I error. (α)

☐ Type-II error:- [False negative]

The error of accepting null hypothesis H_0 when it is false is called Type-II error.

Type-II error is denoted by β

☐ Level of significance:-

denoted by α

The probability of Type-I error is called level of significance.

→ It is denoted by α

→ α could be 1%/5%/10%

☐ P value:- Probability value

P value is the least possible value of α for which we can reject the null hypothesis.

$P > \alpha = 5\% \rightarrow$ rejected [H_0]

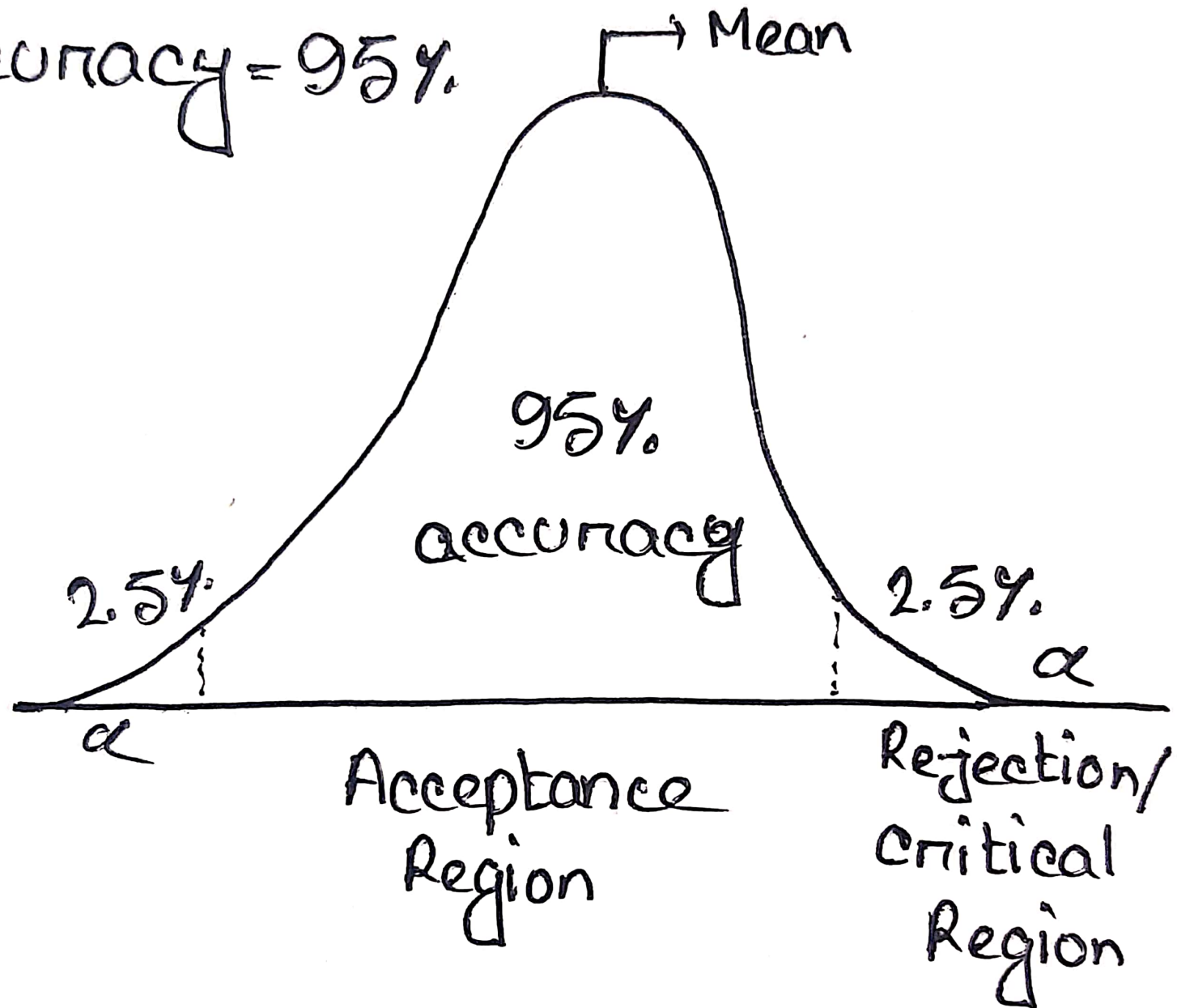
$P < \alpha = 5\% \rightarrow$ accepted [H_0]

df = degree of freedom. → স্বাধীনতার মাত্রা

$$df = (r-1) \times (c-1)$$

$$\alpha = 5\% = 0.05$$

$$\text{accuracy} = 95\%$$



Comment on P value:-

1. If $P < 1\%$; It is highly statistical significant.
2. If $P < 5\%$; It is significant.
3. If $P > 10\%$; It is not significant.

Commonly used Test statistic:-

1. The normal test (Z-test) $\rightarrow$ [large sample size, greater than 30]
2. The 't' test $\rightarrow$ [Small sample, less than 30].
3. Chi-square (χ^2) test.
4. F-test.

Application of Chi-square (χ^2) test:-

1. To test the significance of a specified population variance.
2. To test the goodness of fit of a distribution.
3. To test the independence of attributes.
4. To test the equality of several correlation coefficients.
5. To test the homogeneity of several tests.
6. To test the homogeneity of several population variance.

Subject / Theme :

Application of F-test:-

1. To test the equality of several population means
2. To test the equality of two population variance.
3. To test the significance of regression.
4. To test the significance of multiple correlation coefficient.

Subject / Theme :

Problem-1

Two hundred engineers were interviewed and classified according to their results and job ~~ate~~ satisfaction. The distribution of graduates by results and job satisfaction are given in following contingency table:

Results	Job Satisfaction	
	Yes	No
Excellent	20	70
Good	45	65

Test at 5% level of significance whether there is any relationship between result and job satisfaction.

Note: At 5% level of significance tabulated value of chi-square @ 1df = 3.84.

H₀: There is no relation between results and job satisfaction.

H₁: There is relation between results and job satisfaction.

Result	Job satisfaction		Total	
	Yes	No		
Excellent	20(O ₁₁)	70(O ₁₂)	90	→ R ₁
Good	45(O ₂₁)	65(O ₂₂)	110	→ R ₂
Total	65	135	200	
	↓ C ₁	↓ C ₂	↓ N	

We know,

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$E_{ij} = \frac{R_i \times C_j}{N}$$

O = observed value

E = expected value

i = Row & j = col

$$\therefore E_{11} = \frac{90 \times 65}{200} = 29.25$$

$$\therefore E_{12} = \frac{90 \times 135}{200} = 60.75$$

$$\therefore E_{21} = \frac{110 \times 65}{200} = 35.75$$

$$\therefore E_{22} = \frac{110 \times 135}{200} = 74.25$$

$$\begin{aligned} \therefore \chi_c^2 &= \frac{(20 - 29.25)^2}{29.25} + \frac{(70 - 60.75)^2}{60.75} + \frac{(45 - 35.75)^2}{35.75} + \frac{(65 - 74.25)^2}{74.25} \\ &= 2.93 + 1.41 + 2.39 + 1.15 \\ &= 7.88 \end{aligned}$$

Here, $\alpha = 5\% = 0.05$ & $df = 1$.

$$\text{also, } \chi_c^2 = 3.84$$

$$\chi_c^2 > \chi_E^2$$

Comment: H_0 rejected, There is a relationship between results and job satisfaction.